Practice Quiz: Lecture 6 (Part 1)

Topic: Bandwidth Utilization, FDM, and WDM **Questions:** 16 | **Total Marks:** 20

> **Instructions:** Answer all questions. Write answers clearly. Show all calculation steps and specify units where appropriate. Formulas may be referenced at the end of the paper.

1. What is the fundamental objective of bandwidth utilization in data communication networks?

- ☐ a. To mandate the use of Time Division Multiplexing on all high-speed transmission links.
- ☐ b. To achieve the effective and optimal use of available physical channel bandwidth.
- ☐ c. To enforce purely digital transmission schemes across long-distance analog media.
- ☐ d. To calculate the exact bit-to-signal ratio for baseband signaling.
- ☐ e. To maximize latency and jitter across multi-user access networks.

2. Complete the statement regarding general multiplexing principles:

In a multiplexed system, ---------- input signals share the physical bandwidth of ---------- communication data link.

- ☐ a. 1; n
- ☐ b. n; n
- ☐ c. n; 1
- ☐ d. 1; 1

3. Which transmission signal type is fundamentally supported across each channel of a link running Frequency Division Multiplexing (FDM)?

- ☐ a. Analog transmission
- ☐ b. Pure baseband digital transmission
- ☐ c. Space transmission
- ☐ d. Discrete packet-switched transmission

4. (Calculation) Multiplexing with Guard Bands

Five channels, each with a bandwidth of $100\ \text{kHz}$, are to be multiplexed using FDM. If a guard band of $5\ \text{kHz}$ is required between adjacent channels to prevent cross-talk interference, what is the minimum required total bandwidth of the composite link (in kHz)?

5. (Calculation) Guard Bands with Increased Channels

Eight analog channels, each with a bandwidth of $250\ \text{kHz}$, are multiplexed over a shared medium. Adjacent channels are separated by guard bands of $12\ \text{kHz}$. Calculate the total composite bandwidth required (in kHz).

6. (Calculation) Determining QAM Order from Data and Baud Rates

A transmitter sends digital data at a transmission bit rate of $1.25\ \text{Mbps}$ across an analog channel using Quadrature Amplitude Modulation. If the signaling baud rate is $250\ \text{kbaud/s}$, which M-QAM scheme is implemented?

7. (Calculation) Baud Rate and Modulation

A modem operates at a data rate of $36\ \text{Mbps}$ over a bandpass link. If the modem uses 64-QAM modulation, what is the resulting baud rate (in Mbaud/s)?

8. (Design Problem) FDM Satellite Configuration

Five digital data channels, each transmitting at $8\ \mathrm{Mbps}$, are allocated to an analog satellite channel having a total bandwidth of $10\ \mathrm{MHz}$. Assuming FDM with uniform channel division without guard bands:

a. What is the allocated analog bandwidth per channel?
b. How many bits per cycle/Hertz must each digital channel modulate?
c. Which standard M-QAM constellation achieves this requirement?

9. (Calculation) Required FDM Bandwidth for Digital Channels

An FDM system combines 3 digital data streams. Each stream transmits at $8\ \mathrm{Mbps}$, and 16-QAM is used for modulation. Assuming an ideal bandwidth efficiency of $1\ \mathrm{baud/Hz}$, what is the total minimum bandwidth required for the link in MHz?

10. (Calculation) Composite FDM Bandwidth with Guard Bands

Four digital channels, each operating at $12\ \mathrm{Mbps}$, are modulated using 64-QAM. These channels are multiplexed using FDM, requiring a $200\ \mathrm{kHz}$ guard band between adjacent channels. What is the total composite bandwidth required for the transmission link in MHz?

11. (Conceptual) True or False: Optical WDM Properties

State whether each statement is **True** or **False**:

a. Coarse Wavelength Division Multiplexing (CWDM) supports longer unamplified transmission distances than Dense Wavelength Division Multiplexing (DWDM).
b. Wavelength Division Multiplexing (WDM) is conceptually an analog multiplexing technique identical to FDM, but operating at optical carrier frequencies.

12. At the receiver of an FDM system, which hardware component is primarily responsible for isolating and separating the individual carrier bands from the composite received signal?

- □ a. Analog-to-digital converters (ADC)
- □ b. Bandpass filters (BPF)
- □ c. Frequency up-converters
- □ d. Asynchronous time-slot allocators

13. (Reverse Calculation) Channel Capacity under Bandwidth Constraint

A communication link has an available bandwidth of $1050\ \text{kHz}$. Each voice channel requires a bandwidth of $200\ \text{kHz}$, and adjacent channels must be separated by guard bands of $10\ \text{kHz}$. What is the maximum number of voice channels N that can be multiplexed concurrently?

14. Which spectral optical window is DWDM designed to exploit, primarily due to minimal attenuation and compatibility with Erbium-Doped Fiber Amplifiers (EDFA)?

- □ a. $850\ \text{nm}$ window
- □ b. $1310\ \text{nm}$ window
- □ c. $1550\ \text{nm}$ (C-band) window
- □ d. $650\ \text{nm}$ visible spectrum

15. (Calculation) Voice Spectrum Frequency Mapping

Three voice channels, each occupying a bandwidth of $4\ \text{kHz}$ (from 0 to $4\ \text{kHz}$), are combined using FDM with $1\ \text{kHz}$ guard bands. If the lowest allocated carrier frequency begins at a base frequency of $20\ \text{kHz}$, what is the highest frequency occupied by the composite signal (in kHz)?

16. What is the fundamental operational difference between CWDM and DWDM?

- □ a. CWDM operates over twisted-pair copper, whereas DWDM operates over fiber optics.
- □ b. CWDM provides wider channel spacing (typically $20\ \text{nm}$) and fewer channels, whereas DWDM provides narrower channel spacing (e.g., $0.8\ \text{nm}$) and supports amplification.
- □ c. CWDM uses time slots, whereas DWDM uses frequency modulation.
- □ d. DWDM cannot multiplex optical signals carrying digital data streams.

Answer Key & Detailed Explanations: Lecture 6 (Part 1)

Question 1: b

Explanation: Bandwidth utilization is the wise use of available bandwidth to achieve maximum data transmission efficiency. Multiplexing and spreading are mechanisms designed to serve this primary goal.

Question 2: c (n; 1)

Explanation: In any multiplexer configuration, n input lines share a single (1) common communication link. At the receiving end, a demultiplexer separates the single link back into n output lines.

Question 3: a (Analog transmission)

Explanation: FDM combines analog signals modulated onto different carrier frequencies. Even when the source data is digital, each stream must be modulated onto an analog carrier wave before transmission.

Question 4: 520 kHz

Explanation:

$$\text{Total Bandwidth} = N \times B_{\text{channel}} + (N-1) \times B_{\text{guard}}$$

$$\text{Total Bandwidth} = 5 \times 100\ \text{kHz} + (5-1) \times 5\ \text{kHz} = 500 + 20 = 520\ \text{kHz}$$

Question 5: 2084 kHz

Explanation:

$$\text{Channels' Bandwidth} = 8 \times 250\ \text{kHz} = 2000\ \text{kHz}$$

$$\text{Guard Bands} = (8-1) \times 12\ \text{kHz} = 7 \times 12 = 84\ \text{kHz}$$

$$\text{Total Bandwidth} = 2000\ \text{kHz} + 84\ \text{kHz} = 2084\ \text{kHz}$$

Question 6: 32-QAM

Explanation:

$$\text{Data Rate } R = 1.25 \text{ Mbps} = 1250 \text{ kbps}$$

$$\text{Baud Rate } S = 250 \text{ kbaud/s}$$

$$r = \frac{R}{S} = \frac{1250}{250} = 5 \text{ bits/baud}$$

$$M = 2^r = 2^5 = 32 \implies 32\text{-QAM}$$

Question 7: 6 Mbaud/s

Explanation: For 64-QAM, each symbol carries:

$$r = \log_2(64) = 6 \text{ bits/baud}$$

$$\text{Baud Rate } S = \frac{\text{Bit Rate}}{\text{Bits per symbol}} = \frac{36 \text{ Mbps}}{6 \text{ bits/baud}} = 6 \text{ Mbaud/s}$$

Question 8: (a) 2 MHz | (b) 4 bits/Hz | (c) 16-QAM

Explanation:

a. Allocated Bandwidth per channel:

$$B_{\text{sub}} = \frac{B_{\text{total}}}{N} = \frac{10 \text{ MHz}}{5} = 2 \text{ MHz}$$

b. Spectral Efficiency:

$$\eta = \frac{8 \text{ Mbps}}{2 \text{ MHz}} = 4 \text{ bits/Hz (or bits/baud)}$$

c. Modulation Order:

$$M = 2^4 = 16 \implies 16\text{-QAM}$$

Question 9: 6 MHz

Explanation:

$$\text{Total Data Rate} = 3 \times 8 \text{ Mbps} = 24 \text{ Mbps}$$

$$\text{16-QAM} \implies \log_2(16) = 4 \text{ bits/symbol}$$

$$\text{Baud Rate} = \frac{24 \text{ Mbps}}{4 \text{ bits/baud}} = 6 \text{ Mbaud/s}$$

Assuming standard Nyquist minimum bandwidth ($B = S$), required bandwidth is 6 MHz.

Question 10: 8.6 MHz

Explanation: For each channel:

$$r = \log_2(64) = 6 \text{ bits/baud}$$

$$\text{Baud Rate per channel} = \frac{12 \text{ Mbps}}{6 \text{ bits/baud}} = 2 \text{ Mbaud/s} \implies B_{\text{channel}} = 2 \text{ MHz}$$

For 4 channels:

$$\text{Channels' Bandwidth} = 4 \times 2 \text{ MHz} = 8 \text{ MHz}$$

$$\text{Guard Bands Bandwidth} = (4 - 1) \times 0.2 \text{ MHz} = 3 \times 0.2 = 0.6 \text{ MHz}$$

$$\text{Total Bandwidth} = 8.0 \text{ MHz} + 0.6 \text{ MHz} = 8.6 \text{ MHz}$$

Question 11: (a) False | (b) True

Explanation: (a) **False:** DWDM operates in the 1550 nm window where Erbium-Doped Fiber Amplifiers (EDFAs) can amplify signals without optical-electrical-optical conversion, allowing distances spanning thousands of kilometers. CWDM signals have wider channel spacing across multiple bands and cannot be amplified together using standard optical amplifiers, limiting CWDM to shorter metro distances (typically ≤ 80 km).
(b) **True:** WDM is fundamentally FDM implemented at optical light frequencies (THz range / nanometer wavelengths).

Question 12: b (Bandpass filters)

Explanation: At the receiver, the multiplexed composite signal passes through a bank of analog bandpass filters (BPF), each tuned to pass only the specific carrier frequency band belonging to that designated channel.

Question 13: 5 channels

Explanation:

$$N \times 200 + (N-1) \times 10 \leq 1050$$

$$200N + 10N - 10 \leq 1050$$

$$210N \leq 1060 \implies N \leq \frac{1060}{210} \approx 5.047$$

Since N must be an integer, $N = 5$ channels. Check:
$5 \times 200 + 4 \times 10 = 1000 + 40 = 1040\ \text{kHz} \leq 1050\ \text{kHz}$.

Question 14: c (1550 nm / C-band window)

Explanation: The C-band (around $1530\text{--}1565\ \text{nm}$) offers extremely low silica fiber attenuation ($\approx 0.2\ \text{dB/km}$) and aligns with the operating gain spectrum of EDFAs.

Question 15: 34 kHz

Explanation: Total span occupied:

$$\text{Span} = 3 \times 4\ \text{kHz} + (3-1) \times 1\ \text{kHz} = 12 + 2 = 14\ \text{kHz}$$

$$\text{Highest Frequency} = f_{\text{start}} + \text{Span} = 20\ \text{kHz} + 14\ \text{kHz} = 34\ \text{kHz}$$

Question 16: b

Explanation: CWDM uses coarse wavelength channel spacing (ITU-T G.694.2 defines $20\ \text{nm}$ spacing) accommodating inexpensive uncooled lasers for shorter reach. DWDM utilizes dense spacing ($\leq 0.8\ \text{nm}$ / $100\ \text{GHz}$ or less), accommodating up to 80+ channels with optical amplification for long-haul networks.